Arista Access Node Optical Transport Failure SOP (HEB-ACC-11)

Node: HEB-ACC-11 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_4565

Description

The Optical Transport Failure in the Access Node HEB-ACC-11 is primarily caused by the degradation of the SFP/XFP modules or fiber cuts leading to a loss of signal strength in the receiver (Rx) and/or transmitter (Tx) paths. This failure domain is specific to the Access Role in the Arista network equipment due to the reliance on optical connections for data transmission. As per the Arista Architecture, the Access Node is responsible for receiving and transmitting data packets between the edge switch and the core network. The Optical Transport Failure can be attributed to various factors such as excessive power consumption, temperature variations, or physical damage to the SFP/XFP modules or the fiber optic cabling.

Pre-requisite Checks

- check: Verify the site location and confirm that the issue is specific to the Access Node HEB-ACC-11 at the Hebbal Edge site., action: Dispatch remote hands to the Hebbal Edge site to physically inspect the equipment.

- check: Check the status of the redundant link and confirm whether it is operational and providing an alternate path for data transmission., action: If the redundant link is not operational, restore it to ensure uninterrupted data transmission.

- check: Verify the power supply to the Access Node and confirm that it is within the recommended range., action: Adjust the power supply to the Access Node if necessary.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP Module	High
Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

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Command: show interface all, Expected Output: Interface statistics, including errors and drops, Description: This command provides an overview of the interface statistics, allowing the technician to identify any errors or drops that may indicate an Optical Transport Failure.

Command: show sfp all, Expected Output: SFP/XFP module status, including temperature and power consumption, Description: This command provides detailed information about the SFP/XFP modules, enabling the
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technician to identify any issues related to temperature or power consumption that may contribute to the Optical Transport Failure.

Command: show interface statistics, Expected Output: Detailed interface statistics, including Rx and Tx packets, Description: This command provides detailed statistics about the interface, allowing the technician to analyze the Rx and Tx packets and identify any anomalies that may indicate an Optical Transport Failure.
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Resolution Steps

Step 1: Step: Replace the Degraded SFP/XFP Module, Command: replace sfp interface with , Description: Replace the degraded SFP/XFP module with a new one to resolve the Optical Transport Failure.

Step 2: Step: Repair the Fiber Cut, Command: repair fiber cut , Description: Repair the fiber cut to restore the optical connection and resolve the Optical Transport Failure.

Step 3: Step: Adjust the Rx/Tx Power Settings, Command: adjust rx-tx power settings to , Description: Adjust the Rx/Tx power settings to optimize the signal strength and resolve the Optical Transport Failure.

Post-Check Verification

```
Command: show interface all, Expected Output: Interface statistics, including errors and drops, Description: Verify that the interface statistics have returned to normal, indicating that the Optical Transport Failure has been resolved.

Command: show sfp all, Expected Output: SFP/XFP module status, including temperature and power consumption, Description: Verify that the SFP/XFP module status has returned to normal, indicating that the Optical Transport Failure has been resolved.

Command: show interface statistics, Expected Output: Detailed interface statistics, including Rx and Tx packets, Description: Verify that the interface statistics have returned to normal, indicating that the Optical Transport Failure has been resolved.
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Roll-back Procedures

- Step: Restore the Original SFP/XFP Module, Command: restore sfp interface to , Description: Restore the original SFP/XFP module to its previous state in case the replacement fails.

- Step: Revert the Fiber Repair, Command: revert fiber repair , Description: Revert the fiber repair to its previous state in case the repair fails.

- Step: Restore the Original Rx/Tx Power Settings, Command: restore rx-tx power settings to , Description: Restore the original Rx/Tx power settings to their previous state in case the adjustment fails.